COMPUTER PROGRAMMING

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LAB 5

LAB TITLE

The while Loop in C++

OBJECTIVE

The objectives of this lab are:

- **To understand the concept of the while loop in C++.**
- **To learn how a nested while loop works.**
- **To understand the do...while loop and its behavior.**
- **To write programs that execute repeatedly based on a condition.**
- **To apply loops in solving simple programming problems such as printing numbers and calculating sums.**

2. THEORY

The while Loop

A while loop is a control flow statement used in C++ to execute a block of code repeatedly as long as a given condition remains true.

The condition is checked before executing the loop body. If the condition is true, the statements inside the loop are executed. The process repeats until the condition becomes false.

Syntax

```
   while(condition)
{
 // statements
}
```

Working of while loop

1. **The condition is evaluated.**
2. **If the condition is true, the loop body executes.**
3. **After execution, the condition is checked again.**

The loop continues until the condition becomes false.

Nested while Loop

1. **A nested while loop is a loop inside another loop.**
2. **In this case:**
3. **The outer loop controls how many times the inner loop runs.**
4. **The inner loop completes all its iterations for each iteration of the outer loop.**
5. **Nested loops are useful in problems involving tables, patterns, or matrices.**

Example concept:

```
while(condition1)
{
while(condition2)

   {

      // inner loop statements
}

}
```

The do...while Loop

1. **The do...while loop is similar to the while loop but has one important difference.**
2. **In a do...while loop, the loop body is executed at least once, because the condition is checked after the loop body executes.**

Syntax

```
Do
{
 // statements
}
while(condition);
```

3. SOFTWARE

- DEV-C++ IDE
- C++ Compiler (included in DEV-C++)

Task 1: Display Numbers Using while Loop

In this task, a C++ program was written using a **while loop** to display numbers from **1 to 10**. The loop starts from 1 and continues until the value becomes 10. After printing each number, the counter increases by one. Each number is displayed on a separate line.

```
#include <iostream>
using namespace std;

 int main  ()
{
     int i = 1;
   while(i<=10)
{
cout << i << endl;
     i++;
}
return 0; }
```

Task 2: Display Numbers Using do...while Loop

In this task, the same program was implemented using a **do...while loop**. The loop prints numbers from **1 to 10**. In a do...while loop, the loop body executes first and then the condition is checked. This ensures the loop runs at least once before checking the condition.

```
#include <iostream>
 using namespace std;

   int main()
{     int i = 1;  do
   {
      cout << i << endl;
      i++;
   }
   while(i <= 10);
return 0;
}
```

Task 3: Sum Using while Loop

In this task, a program was written that asks the user to enter a **positive integer**. A **while loop** is used to calculate the sum of numbers from **1 up to the entered number**. The loop keeps adding the numbers and finally displays the total sum.

```
#include <iostream>
using namespace std;
    int main()
{
   int num, i = 1, sum = 0;

cout << "Enter a  positive
integer: ";
cin >> num;
   while(i <= num)
     { sum += i; i++; }

  cout << "Sum = " <<
sum << endl;

return 0; }
```

Conclusion

In this lab we learned how to use **while loops** and **do...while loops** in C++. We also learned how loops help repeat operations efficiently and solve problems like **printing numbers and calculating sums**.